

Subsequence

str $\rightarrow$ "abc"

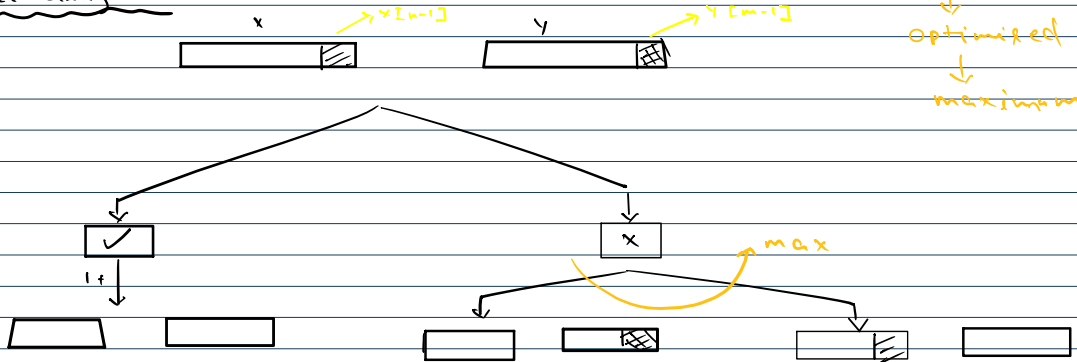
"a", "b", "ab", "ac", "bc", "abc", "c", ""

Longest Common Subsequence

I/p $\rightarrow$ x: a b c d g h
 y: a b e d f h

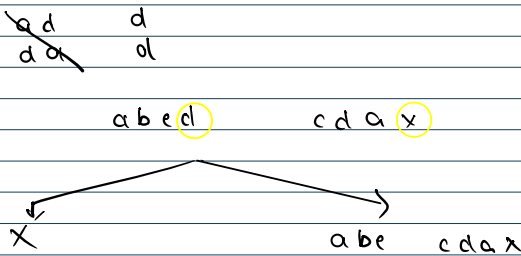
o/p $\rightarrow$ a b d h
 $\hookrightarrow$ length $\rightarrow$ 4

choice diagram



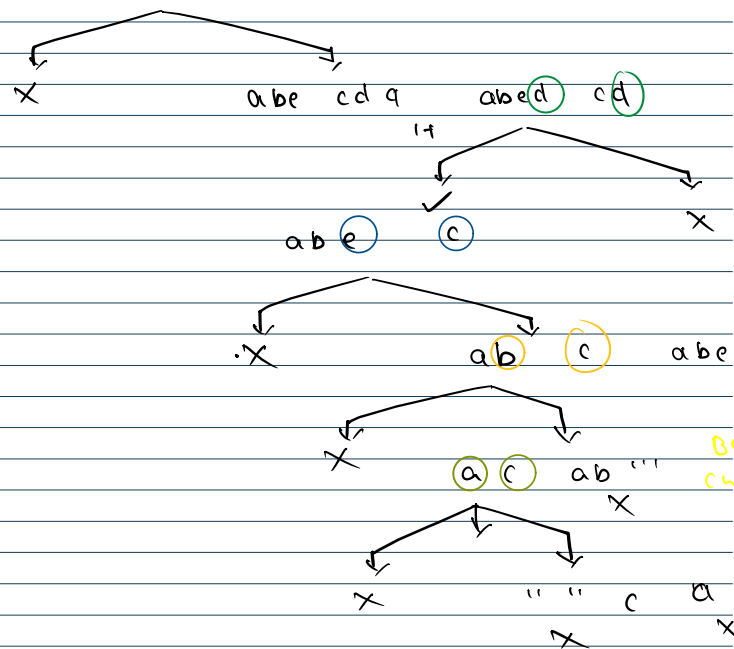
Longest
 $\downarrow$
 optimized
 $\downarrow$
 maximum

x $\rightarrow$ a b e d
 y $\rightarrow$ c d a x



Base case $\Rightarrow$

x.len == 0 or y.len == 0
 $\downarrow$
 return 0



0	-1	0	0	0	-1	-1
-1	1	1	1	1	1	-1
0	1	-1	2	2	2	-1
-1	-1	2	2	2	2	-1

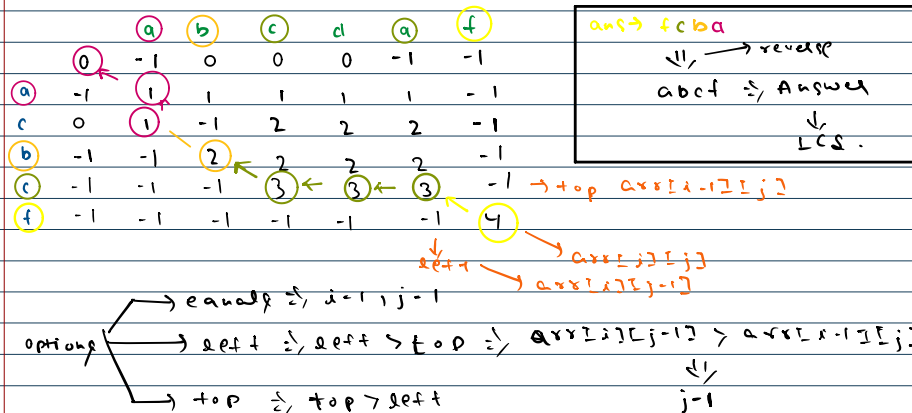


are
d+u

... f

...

0	-1	0	0	0	-1	-1
-1	1	1	1	1	1	-1
0	1	-1	2	2	2	-1
-1	-1	2	2	2	2	-1
-1	-1	-1	3	3	3	-1
-1	-1	-1	-1	-1	-1	4



	a	b	c	d	a	f
a	0	1	1	1	1	1
c	0	1	2	2	2	2
b	0	1	2	2	2	2
c	0	1	2	3	3	3
f	0	1	2	3	3	4

$$dp[i][j] = ?$$

$$\begin{cases} x[i-1] == y[i-1] \\ \Downarrow \\ dp[i][j] = 1 + dp[i-1][j-1] \end{cases}$$

$$dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$$

Minimum no. of insertion and deletion required

slp $\rightarrow$ a : heap
b : pea

a $\rightarrow$ b

$\Downarrow$
count of
insertion
+
deletion

p/heap $\rightarrow$ pea

Insert $\rightarrow$ 1 $\rightarrow$ heap $\rightarrow$ len(b) - LCS $\Rightarrow$ P $\Rightarrow$ ①

Delete $\rightarrow$ 2 $\rightarrow$ heap $\rightarrow$ len(a) - LCS $\Rightarrow$ h.p $\Rightarrow$ ②

Answer $\Rightarrow$ len(a) + len(b) - 2.LCS

